

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Data Warehousing and Data Mining
COURSE CODE: CSA1609

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

Criteria	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I G Madhulatha(192311295)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G Madhulatha

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K	SSE
1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72

E4	55	65	58
E5	93	96	90

Silhouette Scores

Model	Score
K=2	0.48
K=3	0.68
K=4	0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

ANSWERS

Q1. Dataset: Hospital Patient Segmentation

Question

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Answer

K-Means clustering is an unsupervised machine learning technique used to divide data into a specified number of clusters. Patients within the same cluster should have similar characteristics, such as medical expenses and health risk scores.

The given SSE values are:

Number of Clusters (K)	SSE
1	600
2	350
3	220
4	200
5	190

Step 1: Understand SSE

SSE (Sum of Squared Errors) measures how closely the data points are grouped around their cluster centroids.

- A high SSE indicates that the clusters are relatively spread out.
- A low SSE indicates that the data points are closer to their cluster centroids.
- Increasing K generally decreases SSE because more clusters provide more centroids.

However, selecting the largest K is not necessarily the best choice.

Step 2: Identify the elbow point

Observe the decrease in SSE:

- $K = 1 \rightarrow K = 2$: $600 \rightarrow 350$, decrease = 250
- $K = 2 \rightarrow K = 3$: $350 \rightarrow 220$, decrease = 130
- $K = 3 \rightarrow K = 4$: $220 \rightarrow 200$, decrease = 20
- $K = 4 \rightarrow K = 5$: $200 \rightarrow 190$, decrease = 10

The major reduction occurs up to $K = 3$. After $K = 3$, the improvement becomes very small.

Therefore, the elbow point is $K = 3$.

Step 3: Optimal number of clusters

The optimal number of clusters is:

$K = 3$

This means the healthcare organization should divide the patients into three major groups.

A possible interpretation is:

- Cluster 1 – Lower-risk/lower-expense patients
 - Relatively low medical expenses
 - Lower or moderate health risk
 - May require routine healthcare and preventive monitoring.
- Cluster 2 – High-risk patients
 - Higher health risk scores
 - May require regular medical monitoring and early intervention.
- Cluster 3 – High-expense patients
 - Patients generating relatively high medical expenses
 - May need specialized treatment, chronic disease management, or closer financial and healthcare planning.

The actual cluster composition would depend on the complete K-Means calculation using the selected features.

Step 4: Effect on patient segmentation

Selecting $K = 3$ provides a useful balance between simplicity and meaningful segmentation.
If $K = 1$ were selected:

- All patients would be treated as one group.
- Differences between patients would be ignored.
- Personalized healthcare decisions would be difficult.

If $K = 2$ were selected:

- Some important differences between patient groups might be lost.
- Patients with different risk and expense characteristics could be combined.

If $K = 3$ is selected:

- Patients can be divided into more meaningful groups.
- Similar patients can receive similar levels of care.
- Healthcare resources can be allocated more effectively.

If $K = 5$ were selected:

- Every patient in this small dataset could potentially become a very small cluster.
- SSE improves only slightly from 200 to 190.
- The additional complexity does not provide much extra information.

Impact on healthcare decision-making

Using three clusters can support:

- **Risk-based healthcare management**
High-risk patients can receive more frequent monitoring.
- **Resource allocation**
Hospitals can allocate doctors, beds, medicines, and diagnostic facilities according to patient groups.
- **Personalized treatment**
Treatment and follow-up plans can be designed according to the characteristics of each cluster.
- **Cost management**
High medical-expense groups can be analyzed to identify expensive treatment patterns.
- **Preventive healthcare**
Moderate-risk patients can be targeted with preventive programs before their condition becomes severe.

Conclusion

The SSE values show a significant reduction up to $K = 3$, followed by only small improvements. Therefore, 3 clusters is the optimal choice according to the elbow method. This segmentation provides a practical balance between clustering quality and model simplicity and can help the healthcare organization provide more targeted treatment, monitoring, and resource allocation.

Q2. Dataset: E-Commerce Order Analysis

Question

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Answer

Hierarchical clustering is an unsupervised clustering technique that creates a hierarchy of groups. The results are commonly represented using a dendrogram.

The dataset is:

Order ID	Product Category	Order Value (₹)
O1	Electronics	55,000
O2	Clothing	25,000
O3	Electronics	57,000
O4	Furniture	30,000
O5	Clothing	27,000

Step 1: Understand hierarchical clustering

Hierarchical clustering generally starts by treating each observation as an individual cluster.

Initially:

- O1 → Cluster 1
- O2 → Cluster 2
- O3 → Cluster 3
- O4 → Cluster 4
- O5 → Cluster 5

The algorithm calculates similarities or distances between observations and progressively merges the most similar observations. The process continues until all observations belong to one cluster.

Step 2: Identify similar orders

From the given data:

- O1 and O3 are both Electronics orders and have similar values:
 - O1 = ₹55,000
 - O3 = ₹57,000

Therefore, they are highly similar.

Similarly:

- O2 and O5 are both Clothing orders:
 - O2 = ₹25,000
 - O5 = ₹27,000

They are also highly similar.

O4 is a Furniture order with an order value of ₹30,000, so it is different in product category from O2/O5 and substantially different in value from O1/O3.

Step 3: Cluster formation

A likely hierarchical clustering structure is:

First group:

O1 + O3 → Electronics cluster

Second group:

O2 + O5 → Clothing cluster

Third group:

O4 → Furniture cluster

Thus, cutting the dendrogram at an appropriate distance level can produce approximately 3 clusters.

Step 4: Appropriate number of clusters

The appropriate number of clusters can be considered as:

K = 3

The resulting groups can be interpreted as:

Cluster	Orders	Main Characteristic
Cluster 1	O1, O3	High-value Electronics
Cluster 2	O2, O5	Medium-value Clothing
Cluster 3	O4	Furniture order

The exact number should normally be determined by observing the largest meaningful vertical gap in the dendrogram. Since the actual dendrogram image is not supplied here, the three-cluster interpretation is based on the similarities in the given dataset.

Step 5: Significance of the clusters

Cluster 1 – High-value Electronics

Orders:

- O1 = ₹55,000
- O3 = ₹57,000

This cluster represents customers making high-value electronics purchases.

The retailer could:

- Provide premium offers.
- Recommend related electronic products.
- Introduce extended warranties.
- Provide loyalty rewards.

Cluster 2 – Clothing

Orders:

- O2 = ₹25,000
- O5 = ₹27,000

This group represents customers purchasing clothing products with relatively similar order values.
The retailer could:

- Recommend fashion products.
- Offer clothing bundles.
- Provide seasonal discounts.
- Create personalized clothing advertisements.

Cluster 3 – Furniture
Order:

- O4 = ₹30,000

This represents a furniture purchasing pattern.
The retailer could:

- Recommend home furnishing products.
- Offer complementary products.
- Provide delivery and installation promotions.

Step 6: Importance for purchasing behavior

Hierarchical clustering helps the company understand:

- Product preferences
Customers can be grouped according to the types of products they purchase.
- Spending patterns
Similar order values can reveal high-, medium-, and low-value purchasing behavior.
- Targeted marketing
Different clusters can receive different advertisements and offers.
- Product recommendation
Products related to the customer's cluster can be recommended.
- Customer segmentation
The company can understand different purchasing patterns rather than treating all customers identically.

Conclusion

The hierarchical clustering process progressively merges similar orders. Based on the provided data, a meaningful interpretation is three clusters: high-value Electronics, Clothing, and Furniture. These clusters help the retailer identify purchasing patterns and support targeted marketing, personalized recommendations, product bundling, and customer relationship management.

Q3. Dataset: Mobile App User Behavior

Question

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Answer

HAC (Hierarchical Agglomerative Clustering) is a hierarchical clustering approach that starts with individual users as separate clusters and progressively combines similar users.

The dataset is:

User	Daily Usage (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2,000
U2	60	5	5,000
U3	140	22	1,500
U4	50	4	6,000
U5	130	20	1,800

Step 1: Analyze user behavior

There are three important characteristics:

- Daily usage time
- Login frequency
- In-app purchase amount

Users with similar behavior should have similar cluster memberships.

Step 2: Identify behavioral similarities

U1, U3 and U5

Their behavior is:

- U1: 120 minutes, 18 logins, ₹2,000
- U3: 140 minutes, 22 logins, ₹1,500
- U5: 130 minutes, 20 logins, ₹1,800

These users have:

- High daily usage
- High login frequency
- Relatively lower purchase amounts

Therefore, they can be considered a high-engagement/low-to-moderate spending group.

U2 and U4

Their behavior is:

- U2: 60 minutes, 5 logins, ₹5,000
- U4: 50 minutes, 4 logins, ₹6,000

These users have:

- Lower usage time
- Lower login frequency
- Higher purchase amounts

Therefore, they can be considered a low-engagement/high-spending group.

Step 3: Probability-based membership

In soft clustering, a user does not have to belong completely to only one cluster.

Instead, a membership probability can be assigned.

For example, suppose the analysis produces the following illustrative membership values:

User	High-Engagement Cluster	High-Spending Cluster
U1	0.90	0.10
U2	0.20	0.80
U3	0.95	0.05
U4	0.10	0.90
U5	0.92	0.08

Important: These values are illustrative because actual probability values cannot be calculated from the supplied dataset alone without specifying the soft-clustering algorithm and its parameters.

Step 4: Interpret membership values

For U1:

- High-engagement membership = 0.90
- High-spending membership = 0.10

Therefore, U1 is strongly associated with the high-engagement cluster.

For U2:

- High-engagement membership = 0.20
- High-spending membership = 0.80

Therefore, U2 is more likely to belong to the high-spending group.

For U3:

- Membership = 0.95 in high-engagement cluster

This indicates a very strong association with the high-engagement group.

For U4:

- Membership = 0.90 in high-spending cluster

This indicates that U4 is strongly associated with the high-spending group.

For U5:

- Membership = 0.92 in high-engagement cluster

Therefore, U5 is strongly associated with high engagement.

Step 5: Hard clustering vs soft clustering

Hard clustering assigns each user to exactly one cluster.

Example:

U1 → Cluster A

There is no indication of how strongly U1 belongs to that cluster.

Soft clustering gives membership values.

Example:

U1 → 90% Cluster A, 10% Cluster B

This provides more information.

Advantages of soft clustering

1. Handles overlapping behavior

Real customers may not fit into one category.

For example, a user may:

- Spend a lot of time using the app
- Login frequently
- Also make high-value purchases

Soft clustering can represent this mixed behavior.

2. Provides more detailed customer profiles

Instead of simply saying:

"User belongs to Cluster 1"

the company can understand:

"User has a strong association with Cluster 1 but also shows characteristics of Cluster 2."

3. Better personalization

Marketing strategies can be based on membership strength.

For example:

- High engagement → notifications and new-content recommendations.
- High spending → premium offers.
- High membership in both → personalized premium campaigns.

4. Handles uncertain users

Some users may lie between two behavioral groups. Soft clustering can represent this uncertainty instead of forcing the user into a single category.

5. Better customer behavior analysis

Customer behavior changes over time. A user may gradually move from one behavioral group to another. Soft membership values make these transitions easier to observe.

Conclusion

The dataset indicates a strong similarity between U1, U3, and U5, while U2 and U4 show another distinct behavior pattern. Probability-based membership allows users to have varying degrees of association with different clusters. Therefore, soft clustering is more flexible than hard clustering for customer behavior analysis, especially when user behavior overlaps or changes over time.

Q4. Dataset: Agricultural Crop Monitoring

Question

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Answer

The dataset contains:

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

The three features have very different numerical scales.

- Rainfall: approximately 280–520
- Soil fertility: approximately 3–9
- Crop yield: approximately 65–90

Step 1: Clustering before normalization

Suppose a distance-based clustering algorithm such as K-Means is applied directly to the original data.

A common distance measure is Euclidean distance:

The rainfall values range over hundreds, whereas soil fertility values range only from 3 to 9.

Therefore, rainfall can contribute much more to the calculated distance than soil fertility.

Example

Consider F1 and F4:

F1:

- Rainfall = 300
- Fertility = 8
- Yield = 85

F4:

- Rainfall = 280
- Fertility = 9
- Yield = 90

Differences:

- Rainfall difference = 20
- Fertility difference = 1
- Yield difference = 5

Squared differences:

Total:

Here, the rainfall difference contributes 400 out of 426 to the squared distance.

Therefore, the clustering algorithm may mainly consider rainfall while giving relatively little importance to soil fertility.

Step 2: Problems before normalization

Without normalization:

- Rainfall dominates the distance calculation.
- Soil fertility has less influence.
- Crop yield may also have less influence.
- Clusters may primarily represent rainfall levels rather than overall agricultural conditions.

For example, farms with similar rainfall could be grouped together even if their soil fertility and crop yield are significantly different.

Step 3: Apply normalization

Normalization converts the variables to comparable scales.

A common method is Min-Max normalization:

This transforms values approximately into the range:

For rainfall:

- Minimum = 280
- Maximum = 520

For soil fertility:

- Minimum = 3
- Maximum = 9

For crop yield:

- Minimum = 65
- Maximum = 90

After normalization, the three variables have comparable ranges.

Step 4: Normalized values

Using Min-Max normalization:

Farm	Rainfall	Fertility	Yield
F1	0.083	0.833	0.800
F2	0.917	0.167	0.200
F3	0.500	0.500	0.600

F4	0.000	1.000	1.000
F5	1.000	0.000	0.000

Now each feature has a comparable numerical scale.

Step 5: Interpretation after normalization

After normalization, clustering considers:

- Rainfall
- Soil fertility
- Crop yield

more fairly.

For example:

F1 and F4

- Both have high soil fertility.
- Both have high crop yield.
- Their rainfall is also relatively low.

Thus, they can form a meaningful agricultural-performance group.

F2 and F5

- Both have high rainfall.
- Both have low soil fertility.
- Both have lower crop yield.

Thus, they are likely to form another meaningful group.

F3

- Has moderate rainfall.
- Has moderate fertility.
- Has moderate-to-high yield.

It may form a middle or intermediate cluster.

Possible clustering interpretation

Cluster	Farms	Agricultural characteristics
Cluster 1	F1, F4	Lower/moderate rainfall, high fertility, high yield
Cluster 2	F2, F5	High rainfall, low fertility, low yield
Cluster 3	F3	Moderate characteristics

The exact clusters depend on the selected clustering algorithm, distance metric, and number of clusters.

Step 6: Importance of normalization

Normalization improves clustering because:

1. Prevents scale dominance

A feature with large numerical values does not automatically dominate the clustering process.

2. Gives equal opportunity to features

Rainfall, soil fertility, and crop yield can all contribute meaningfully.

3. Produces meaningful clusters

Clusters represent overall agricultural conditions rather than merely one numerical variable.

4. Improves comparison

Farms can be compared based on multiple characteristics simultaneously.

5. Supports better agricultural decisions

The research center can identify:

- High-performing farms
- Farms with low fertility
- Farms affected by excessive rainfall
- Farms requiring irrigation or soil improvement

Step 7: Impact on agricultural decision-making

The normalized clusters can help agricultural organizations:

- Recommend suitable crops.
- Identify farms requiring fertilizer.
- Improve irrigation planning.
- Study rainfall and yield relationships.
- Identify high-performing farming conditions.
- Develop farm-specific agricultural recommendations.

Conclusion

Before normalization, rainfall can dominate the distance calculation because it has a much larger numerical scale than soil fertility. After normalization, all features contribute more comparably to clustering. Therefore, normalization improves the reliability and interpretability of agricultural clustering, resulting in more meaningful farm segmentation and better agricultural decision-making.

Q5. Dataset: Employee Performance Evaluation

Question

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Answer

The employee dataset is:

Employee	Productivity	Attendance (%)	Skill Assessment
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

The silhouette scores are:

Number of Clusters	Silhouette Score
K = 2	0.48
K = 3	0.68
K = 4	0.52

Step 1: Understand silhouette score

The silhouette score measures how well each data point fits within its assigned cluster compared with other clusters.

The score generally ranges from:

Interpretation:

- Close to +1 → strong clustering
- Around 0 → overlapping clusters
- Negative value → possible incorrect cluster assignment

Therefore, a higher silhouette score indicates better-defined clusters.

Step 2: Compare the models

K = 2

Silhouette score:

This indicates a moderate clustering structure.

The two clusters are reasonably separated, but some employee groups may still be mixed together.

K = 3

Silhouette score:

This is the highest score among the three models.

Therefore, K = 3 produces the most well-separated and internally cohesive clusters.

K = 4

Silhouette score:

The score is better than K = 2 but significantly lower than K = 3.

This suggests that four clusters may be unnecessarily dividing the employee population.

Step 3: Select the optimal number of clusters

Compare:

The maximum silhouette score is:

Therefore:

is the most suitable number of clusters.

Step 4: Possible employee segmentation

Based on the values, a three-cluster solution could conceptually represent:

Cluster 1 – High-performing employees

Examples likely include:

- E1
- E5

Characteristics:

- High productivity
- High attendance
- High skill scores

These employees demonstrate strong overall performance.

Cluster 2 – Moderate-performing employees

A likely member:

- E3

Characteristics:

- Moderate-to-high productivity
- Good attendance
- Moderate skill score

These employees may have good performance but still have opportunities for improvement.

Cluster 3 – Employees requiring improvement

Possible members:

- E2
- E4

Characteristics:

- Lower productivity
- Lower attendance
- Lower skill assessment scores

These employees may benefit from additional training, mentoring, or performance support.

The exact employee-to-cluster assignments should be obtained from the actual clustering algorithm rather than inferred solely from the table.

Step 5: Why $K = 3$ is better

Choosing $K = 3$ provides a useful balance.

With $K = 2$:

- Employee differences may be oversimplified.
- Moderate-performing employees could be grouped with either high- or low-performing employees.
- The silhouette score is only 0.48.

With $K = 3$:

- High, moderate, and lower performance groups can be identified.
- Clusters are better separated.
- Silhouette score is highest at 0.68.

With $K = 4$:

- The employee population may be divided too finely.
- Some clusters may contain very similar employees.
- The silhouette score falls to 0.52.

Step 6: Contribution to employee performance evaluation

The selected three-cluster model can help the organization in several ways.

1. Identify high performers

Employees in the high-performance cluster can be considered for:

- Rewards
- Promotions
- Leadership responsibilities
- Recognition programs

2. Identify employees needing support

Employees in the lower-performance cluster can receive:

- Training
- Mentoring
- Skill-development programs
- Attendance improvement plans

3. Personalized development

Employees in the moderate cluster may require targeted improvement rather than intensive intervention.

4. Objective evaluation

Instead of evaluating employees only individually, clustering identifies groups with similar performance characteristics.

5. Better HR decision-making

The company can use clusters to support:

- Training allocation
- Performance management
- Career development
- Incentive planning
- Workforce planning

Conclusion

The silhouette scores clearly indicate that $K = 3$ is the optimal clustering model, because it has the highest silhouette score of 0.68. The three-cluster model provides better separation and cohesion than $K = 2$ or $K = 4$. It can help the organization distinguish high-performing, moderate-performing, and employees requiring improvement, enabling more effective and targeted employee performance management.

Quick Final Answers

Question	Method/Measure	Best Choice	Main Reason
Q1	K-Means + Elbow Method	$K = 3$	SSE drops significantly until $K=3$ and then decreases only slightly
Q2	Hierarchical Clustering + Dendrogram	3 clusters	Electronics, Clothing, and Furniture show meaningful grouping
Q3	HAC + Soft/Probability Membership	Soft clustering	Represents overlapping and uncertain user behavior
Q4	Clustering + Normalization	Normalized data	Prevents rainfall from dominating distance calculations
Q5	Silhouette Analysis	$K = 3$	Highest silhouette score = 0.68

Important exam points to remember

- Elbow Method: Choose the point where further increases in K produce only small SSE improvements.
- Hierarchical Clustering: Use the dendrogram and cut it at a meaningful distance level.
- Soft Clustering: Gives membership probabilities/degrees of belonging instead of one fixed cluster.
- Normalization: Places variables on comparable scales and prevents large-scale variables from dominating distance calculations.
- Silhouette Score: Higher is better; it indicates better separation and cohesion of clusters.